

# Scoring and leaderboard metrics

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## Abstract

The board does not rank on profit. It ranks on a *doubly normalised* score: each block's profit is first divided by that block's hidden maximum value, and then standardised against the other nineteen bots that played the same block. This document defines that score and every figure reported beside it, gives the formula for each, and says which of them are ranking quantities and which are diagnostics. Section 7 covers the mock auctions, which run the same machinery but do not feed the final standing.

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# 1 Notation

One *game* is a group of  $N = 20$  bots playing  $T = 2000$  rounds, split into  $B = 4$  blocks of  $L = 500$ . One *iteration* is a fresh draw of groups across the whole field; a *showdown* is a set of iterations played back to back and published as one leaderboard.

| Symbol                                         | Meaning                                  | Where it lives                     |
|------------------------------------------------|------------------------------------------|------------------------------------|
| $i$                                            | a bot (index within its group)           | <code>key</code>                   |
| $b \in \{1, \dots, B\}$                        | a block of $L$ rounds                    | <code>BlockSummary.block</code>    |
| $t$                                            | a round, $1 \leq t \leq T$               | <code>obs["round"]</code>          |
| $[m_b, M_b]$                                   | the block's hidden value bounds          | <code>block_bounds</code> (secret) |
| $x_{i,t}$                                      | bot $i$ 's private value in round $t$    | <code>obs["x"]</code>              |
| $n_t$                                          | players still solvent in round $t$       | <code>obs["num_players"]</code>    |
| $X_t$                                          | max value over the <i>active</i> players | <code>max_value</code>             |
| $b_{(1)} \geq b_{(2)} \geq \dots$              | the round's bids, sorted                 | <code>sorted_bids</code>           |
| $C_{i,t}$                                      | bot $i$ 's capital after round $t$       | <code>Player.capital</code>        |
| $C_{i,b}^{\text{start}}, C_{i,b}^{\text{end}}$ | capital at the block's ends              | <code>BlockSummary</code>          |

$X_t$  is the maximum over the players *active that round*. A bankrupt bot draws no value, so it contributes nothing to  $X_t$  and is not counted in  $n_t$ . That is why  $n_t$  is handed to every bot: as the field thins,  $\mathbb{E}[X_t]$  falls.

## 2 What happens inside a block

### 2.1 Values

Every active bot independently draws

$$x_{i,t} \sim \mathcal{U}[m_b, M_b], \quad (1)$$

with  $[m_b, M_b]$  fixed within a block and redrawn at every boundary. Neither bound, nor the block index, nor the fact that a boundary has just happened, is ever reported to a bot. Detecting the regime change is part of the problem.

### 2.2 Capital

Capital is redrawn independently for every bot at the start of every block:

$$\kappa \sim \mathcal{U}[0.5, 2.5], \quad u \sim \mathcal{U}[-10, 10], \quad C_{i,b}^{\text{start}} = \max(\kappa M_b + u, 0.05 M_b). \quad (2)$$

What a bot finished the previous block with does *not* carry over. The floor  $0.05 M_b$  only binds when  $M_b$  is small; it exists to rule out a zero or negative starting capital.

Two consequences that shape the whole score. First, within a block some bots start rich and some start poor, so a strategy that behaves identically on a half- $M_b$  bankroll and a  $2.5M_b$  one is being measured on exactly that. Second, because capital resets, the four blocks are four *independent* tests: a bot cannot bank a good block and coast.

### 2.3 Legality and bankruptcy

The legal bid ceiling is the bot's own capital. A bid above it, negative, NaN, infinite, not a number, or not returned inside the per-round time limit is filed as a bid of 0 for that round.

Capital updates as  $C_{i,t} = C_{i,t-1} + \text{payoff}$ . A bot with  $C_{i,t} \leq 0$  is eliminated *for the remainder of that block* and returns at the next boundary on a fresh draw from (2). Bankruptcy is therefore never fatal, but it forfeits the rest of the block, which is expensive in exactly the way Section 4.6 describes.

## 2.4 Payoffs

With  $X = X_t$  and  $b_1 \geq b_2 \geq \dots$  the sorted bids of the round:

### Variation 1 — private value, first price

The winner takes  $x_i - b_1$ , using *the winner's own* value; everyone else zero.

### Variation 2 — common value, first price

The winner takes  $X - b_1$ ; everyone else zero.

Variations 3 and 4 are released after mock auction 1 and are not described here. Nothing below depends on which variation is being played: the scoring is identical for all four.

**Ties.** Variations 1 and 2 let *every* bidder within  $10^{-9}$  of the top bid win, each collecting the full payoff.

## 3 The ranking quantity

### 3.1 Step 1 — normalised block profit $\pi$

Raw profit is not comparable across blocks: a block with  $M_b = 500$  pays out roughly fifty times one with  $M_b = 10$ , and inside a single block bots start from different capitals. So profit is measured in units of the block's own hidden maximum:

$$\pi_{i,b} = \frac{C_{i,b}^{\text{end}} - C_{i,b}^{\text{start}}}{M_b} \quad (3)$$

with  $\pi_{i,b} := 0$  in the degenerate case  $M_b \leq 0$ . A bot that went bankrupt ends the block on zero capital, so its  $\pi_{i,b} = -C_{i,b}^{\text{start}}/M_b$ , which by (2) is at worst about  $-2.5$ .

$\pi$  is the only quantity in the whole system with an absolute meaning. Everything downstream is relative.  $\pi_{i,b} > 0$  means the strategy made money that block;  $\pi_{i,b} < 0$  means it lost money, however well it placed.

### 3.2 Step 2 — standardisation within the group

Let  $\mathcal{G}$  be the group that played block  $b$  together,  $|\mathcal{G}| = N$ . Write

$$\mu_b = \frac{1}{N} \sum_{j \in \mathcal{G}} \pi_{j,b}, \quad \sigma_b = \sqrt{\frac{1}{N} \sum_{j \in \mathcal{G}} (\pi_{j,b} - \mu_b)^2} \quad (4)$$

(the *population* standard deviation — `statistics.pstdev` — not the sample one; the twenty bots are the whole population of that block, not a sample from it). Then

$$z_{i,b} = \text{clip}\left(\frac{\pi_{i,b} - \mu_b}{\sigma_b}, -3, +3\right), \quad \boxed{P_{i,b} = 50 + 15 z_{i,b}} \quad (5)$$

so  $P_{i,b} \in [5, 95]$ , centred on 50.

Two edge cases are defined explicitly. If  $\sigma_b = 0$  — every bot in the group scored identically — then  $z_{i,b} := 0$  and everyone gets 50; nobody is separated by a block that separated nobody. If the group has a single member, that member gets 50.

The  $\pm 3$  clip is not cosmetic. Without it a single block in which one bot happened to draw a huge  $\kappa$  and win a run of rounds could contribute several hundred points and decide the competition on variance rather than skill. Clipping caps any one block's contribution at 95 and floors it at 5.

### 3.3 Step 3 — iteration score and total score

$$S_i^{\text{iter}} = \sum_{b=1}^B P_{i,b}, \quad \boxed{\text{score}_i = \sum_{\text{iterations } r} S_{i,r}^{\text{iter}}} \quad (6)$$

**score is the ranking quantity, and the only one.** With  $B = 4$  blocks per iteration and points centred on 50, an average bot scores about 200 per iteration; over a three-iteration showdown, about 600.

Because **score** is a *sum*, a bot that played fewer iterations than the rest scores lower purely for that. This is what **mean\_block\_points** exists to correct, and it is the documented tie-break (Section 5).

## 4 Every column on the board

Below,  $\mathcal{B}_i$  is the multiset of all blocks bot  $i$  played across the whole showdown ( $|\mathcal{B}_i| = \text{blocks}$ ) and  $\mathcal{G}_i$  the set of games ( $|\mathcal{G}_i| = \text{games}$ ). All of these are computed in **harness/evaluate.py:aggregate**.

### 4.1 score — Total score

Equation (6). Higher is better. This is what the board sorts on.

### 4.2 mean\_block\_points — Mean block score

$$\bar{P}_i = \frac{1}{|\mathcal{B}_i|} \sum_{b \in \mathcal{B}_i} P_{i,b} \in [5, 95] \quad (7)$$

**score** normalised by how much a bot actually played. It is the tie-break, and it is the honest number to quote when two bots played different numbers of games. Read it against 50: above means the bot beat its group’s average block, below means it did not.

### 4.3 mean\_normalised\_profit — Mean $\pi$

$$\boxed{\bar{\pi}_i = \frac{1}{|\mathcal{B}_i|} \sum_{b \in \mathcal{B}_i} \pi_{i,b}} \quad (8)$$

**The absolute-profitability check.** Standardisation makes  $P$  a purely relative statement: in a group where every bot loses money, somebody still scores 95.  $\bar{\pi}$  is the answer to “but did it actually make money?” — expressed in units of  $M_b$ , so  $\bar{\pi} = +0.4$  means the strategy earned, on average, forty percent of the block’s maximum value.

A bot can rank first on **score** with  $\bar{\pi} < 0$ . That is not a contradiction and not a bug: it means the bot lost the least in a field that all lost. It is, however, exactly the situation a report should notice.

### 4.4 normalised\_profit\_spread — Consistency

$$\boxed{s_i = \sqrt{\frac{1}{|\mathcal{B}_i|} \sum_{b \in \mathcal{B}_i} (\pi_{i,b} - \bar{\pi}_i)^2}} \quad (9)$$

The population standard deviation of  $\pi$  across *all* blocks the bot played, in every game. **Lower is better**, and the board sorts this column ascending for that reason.

This is the column that separates a strategy from a lottery ticket. Two bots can share a  $\bar{\pi}$  of  $+0.3$  with one earning  $(0.3, 0.3, 0.3, 0.3)$  and the other  $(1.5, -0.4, 0.3, -0.2)$ ; they are not the same bot, and the second one is one bad draw from bankruptcy. Note also that a large  $s_i$  is often

the signature of a bot that has *not* detected the block boundary: it is tuned to one regime and is being carried by the blocks that happen to match it.

The four blocks in an iteration differ deliberately in both scale and width, so  $s_i$  measures robustness to the regime change, not just round-to-round noise.

#### 4.5 worst\_normalised\_profit — Worst block $\pi$

$$\pi_i^{\min} = \min_{b \in \mathcal{B}_i} \pi_{i,b} \quad (10)$$

The tail, reported directly.  $s_i$  can be dragged up by an unusually *good* block, which is not a risk;  $\pi^{\min}$  cannot. A bot whose  $\pi^{\min}$  approaches  $-\kappa$  (Section 2.2) lost its entire starting capital in some block, which is the same statement as `survival_rate`  $< 1$ .

#### 4.6 survival\_rate — Survival

$$\rho_i = \frac{\#\{b \in \mathcal{B}_i : \text{not bankrupt at the end of } b\}}{|\mathcal{B}_i|} \quad (11)$$

The fraction of blocks the bot finished solvent; 1 when it never went broke. A block counts as survived when  $C_{i,b}^{\text{end}} > 0$  at the block's last round — equivalently, when the bot was never eliminated during it.

Survival is a *diagnostic, not a penalty term*: it appears nowhere in (6). It does not need to. Going bankrupt forfeits the remainder of the block, so those rounds earn nothing while the rest of the group keeps compounding, which drives  $\pi_{i,b}$  to its floor and  $P_{i,b}$  towards 5. The scoring penalises bankruptcy through  $\pi$ ;  $\rho_i$  just tells you it happened, and how often.

Read  $\rho_i$  together with  $\bar{\pi}_i$ . A bot with  $\bar{\pi} > 0$  and  $\rho < 1$  is one that funds good blocks with catastrophic ones — the profile most likely to collapse when the hidden bounds change to ones it has never seen.

#### 4.7 Raw profit columns

Reported for context, and never used to rank. With  $\Delta_{i,g}$  the net profit of bot  $i$  in game  $g$  (the sum of  $C^{\text{end}} - C^{\text{start}}$  over that game's blocks):

$$\text{mean\_net\_profit} = \frac{1}{|\mathcal{G}_i|} \sum_g \Delta_{i,g}, \quad \text{std\_net\_profit} = \sqrt{\frac{1}{|\mathcal{G}_i|} \sum_g (\Delta_{i,g} - \bar{\Delta}_i)^2}, \quad (12)$$

$$\text{best\_net\_profit} = \max_g \Delta_{i,g}, \quad \text{worst\_net\_profit} = \min_g \Delta_{i,g}, \quad (13)$$

and `mean_final_capital` is the mean of  $C^{\text{end}}$  over the last block of each game. These are in raw currency units and are therefore dominated by whichever blocks happened to have the largest  $M_b$ ; that is precisely why they do not rank anybody.

#### 4.8 Counters

| Column                | Definition                                                                          | Note                                |
|-----------------------|-------------------------------------------------------------------------------------|-------------------------------------|
| <code>games</code>    | $ \mathcal{G}_i $                                                                   | group-games played this showdown    |
| <code>blocks</code>   | $ \mathcal{B}_i $                                                                   | normally $B \times \text{games}$    |
| <code>wins</code>     | rounds finishing rank 1                                                             | in V1/V2 every tied top bidder wins |
| <code>errors</code>   | <code>get_bid</code> raised, or returned<br>a non-number, NaN, $\pm\infty$ or $< 0$ | each filed as a bid of 0            |
| <code>timeouts</code> | rounds over the time limit                                                          | each filed as a bid of 0            |

An over-capital bid is *not* counted as an error: it is legal to attempt and is simply filed as 0. Only malformed returns are counted, which is why a bot can bleed capital through zero-bids with `errors` sitting at 0.

`wins` is a diagnostic and a deliberately misleading one if read alone: in variation 1 a bot that wins every round by overbidding has a large `wins` and a deeply negative  $\pi$ .

## 4.9 Disqualification

`disqualified` is set by the sandbox, not by the scoring: a policy violation, a memory kill, or repeated failure to return. A disqualified bot keeps its computed row — the numbers stay visible — but always sorts last, below every bot that finished legally.

## 5 Ranking and tie-breaks

Within each variation, rows are ordered by the key

$$\left( \mathbb{K}[\text{disqualified}], \text{--score}, \text{--mean\_block\_points} \right) \quad (14)$$

ascending. In words:

1. every legal bot outranks every disqualified one;
2. then by total score, highest first;
3. ties broken by mean block score, highest first — which is what stops a bot that played fewer games from being punished for the scheduler’s arithmetic rather than its own play.

The four variations are ranked on four separate boards. There is no combined score across variations; a bot entered in one variation is not comparable to a bot entered in another, because  $\pi$  is standardised against a different field.

## 6 The tournament

A single 20-bot arrangement is far too noisy to rank on — who you are grouped with matters as much as how you play. So each variation is played over several iterations.

| Iteration | Grouping           | What it does                                                       |
|-----------|--------------------|--------------------------------------------------------------------|
| 1         | random             | the field shuffled into groups of 20 under seed $S_1$              |
| 2         | random, fresh seed | reshuffled independently: new distributions, capitals, opponents   |
| 3         | strength-balanced  | snake-seeded by cumulative points (below)                          |
| 4–5       | finals             | the top <code>finals_size</code> by cumulative score, head to head |

**Snake seeding.** Bots are sorted by cumulative score and dealt into  $G$  groups in a serpentine: ranks  $1 \dots G$  go to groups  $1 \dots G$ , ranks  $G+1 \dots 2G$  go back down  $G \dots 1$ , and so on. Formally, the bot at sorted position  $p$  (0-based) with  $(r, c) = \text{divmod}(p, G)$  lands in group

$$g(p) = \begin{cases} c & r \text{ even} \\ G - 1 - c & r \text{ odd.} \end{cases} \quad (15)$$

This *balances* groups rather than segregating them: every group ends up with roughly the same average strength, which is what makes standardised scores comparable across groups.

**Remainders.** Under random grouping, a trailing group smaller than half of  $N$  is folded into the previous group rather than left to play mostly against filler bots. Snake seeding needs no such rule: it deals every bot into an existing group by construction.

**Filler bots.** A group short of  $N$  players is topped up with the three published sample bots so that the highest-bid race is a real one. Fillers play for real but never appear on a leaderboard and never contribute a row.

**Independence.** Each iteration uses its own block schedule  $\{[m_b, M_b]\}$ , not merely the same schedule under a new seed, so iteration 2 is a genuinely fresh test rather than a re-roll of the same one. Groups within an iteration are independent games and are run in parallel.

Final ranking is the sum of the two finals iterations, tie-broken by the qualification total.

**How this maps onto a run.** grouping is set *per iteration*, so the table above is one run: ["random", "random", "balanced", "finals", "finals"]. Iterations play in order, and each one seeds on the standing after the ones before it — a balanced iteration snake-seeds on iterations 1–2’s points, and a finals iteration cuts the field on everything above it. A run whose iterations are all **random** needs no standing at all and does not consult one.

A run may also be given a single mode, which then applies to every iteration; a schedule shorter than **iterations** holds its last entry. Where a run does need a standing it did not produce itself — the first balanced run of a fresh board — it takes the last finished showdown *of its own kind* (Section 7), never whichever board happened to finish most recently.

## 7 Mock auctions

Two mock showdowns run before the deadline. They use *the same engine, the same sandbox, the same scoring and the same aggregation* as the graded run — there is no separate code path — and differ in settings, in what is done with the result, and in one recorded fact: every showdown is stamped with what it was for.

| Kind            | What it is                                     | Board                      |
|-----------------|------------------------------------------------|----------------------------|
| <b>practice</b> | the clock, every <code>interval_minutes</code> | rewritten by the next tick |
| <b>mock</b>     | an announced mock auction                      | archived and published     |
| <b>final</b>    | the run after the deadline                     | the result                 |

The stamp is not a label. A **balanced** or **finals** iteration seeds on a standing, and where that standing comes from decides who plays whom: seeding takes the last finished run *of the same kind*, so a mock is seeded by the previous mock and never by a practice run against throwaway bounds. It is also what keeps a mock board reachable after the two-hourly clock has replaced the live one.

|                           | Mock auction 1                       | Mock auction 2 | Final evaluation                     |
|---------------------------|--------------------------------------|----------------|--------------------------------------|
| Variations                | 1, 2                                 | all released   | all released                         |
| Submissions               | optional                             | optional       | required                             |
| Grouping                  | random                               | random         | random $\times 2$ , balanced, finals |
| Iterations                | reduced                              | reduced        | 3 qualification + 2 finals           |
| Bounds                    | throwaway                            | throwaway      | the real hidden schedule             |
| Counts towards the result | no                                   | no             | yes                                  |
| Also does                 | releases variations 3 and 4 (mock 1) |                |                                      |

Four things follow, and they are the whole point of running them.

1. **A mock score is not a partial final score.** Nothing carries over. The graded run starts from an empty board, on block bounds and seeds that have never been published, against whatever field submits by the deadline. A mock rank is a measurement, not a position.

2. **The hidden bounds are different, deliberately.** The mocks are run on a throwaway schedule. A bot tuned to the  $\pi$  it earned in a mock is tuned to a distribution that will not appear again — which is the same trap the local runner warns about, one level up.
3. **The per-block table is the deliverable, not the rank.** Each participant is sent their  $(C_b^{\text{start}}, C_b^{\text{end}}, M_b, \pi_b, P_b, \text{bankrupt})$  row for every block. That is where a bot's failure mode is legible: a  $\pi_b$  that collapses in exactly one block is a bot that has not detected the boundary; a low  $P_b$  against a healthy  $\pi_b$  is a bot that is profitable but being out-played; a bankruptcy tells you which block and which round.
4. **Mock 1 is also the release gate.** The last two variations are published the moment mock auction 1's results are, so their mock field necessarily has fewer entrants and less-developed bots than the final one. Read a mock 2 rank in a late-released variation with that in mind: the standardisation in (5) is against *that* field, and a thin field of under-tuned bots makes  $\sigma_b$  small and  $z$  — and so the whole board — unusually sensitive.

Operationally, a mock run is an ordinary showdown triggered from the admin console and stamped **mock**; the practice board that rewrites itself every two hours is the same thing again, stamped **practice**, at a smaller size. Nothing on either is graded. The only run that is graded is the one after the final deadline, on settings that are not published until it has finished.

## 8 Reading the board: three worked cases

**A bot ranked 3rd with  $\bar{\pi} = -0.15$ .** It is losing money in absolute terms and still beating most of its group. In V1 or V2 that says the whole field is overbidding and the auction is dissipating its own surplus. Check  $\rho_i$ : if it is 1, the bot is simply being conservative in a bloodbath, which is a perfectly good strategy.

**A bot with  $\bar{P} = 62$  and  $s = 1.1$ .** Strong on average, wildly inconsistent. With four blocks per iteration, an  $s$  of 1.1 against a  $\bar{\pi}$  of a few tenths means individual blocks are swinging by more than the block's own maximum value. Look at  $\pi^{\min}$  and  $\rho$ : this is usually a bot that is excellent in one regime and bankrupt in another, and it will not survive a fresh block schedule.

**A bot with  $\rho = 0.75$  and rank 1.** It went bankrupt in a quarter of its blocks and still tops the board, so the blocks it survived were dominant enough to absorb three  $P$ -values near the floor. That is a real result, but it is a  $3\sigma$ -clipped one: the clip in (5) caps how much its good blocks can earn while its bad ones sit at 5. It is close to the point where one more bad draw flips the ranking, and it is the first bot to read by hand.